The flower-to-seed investigation

Teacher guide and worked answers

Describe pollination, fertilisation and seed formation as different connected processes.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C30

Describe reproduction in plants (pollination, seed formation).

New outcome; flowering-plant reproduction sequence. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Identify flower, stem and leaf.

Know that reproduction produces new individuals.

Success evidence

Trace pollen from anther to stigma.

Distinguish pollen transfer from fusion of gametes.

Explain that a fertilised ovule develops into a seed.

Equipment

Shared orchard evidence, chosen route, pencils; optional paper pollen tokens.

Preparation

Use diagrams or paper tokens, not real pollen.

Explain that our example follows ordinary sexual reproduction in a flowering plant.

Practical care

No pollen blowing, plant tasting or flower dissection is required; paper route avoids pollen allergy exposure.

Ready to teach alternative

Trace paper pollen and stage cards using the supplied observations. No live plant, internet or completed growing trial is needed.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A pollen delivery mystery
2	Retrieval	2-5	Recover the plant words
3	I do	5-13	I trace the correct delivery destination

Slide	Stage	Minutes	Focus
4	I do	Within stage	I distinguish the event inside the ovule
5	I do	Within stage	I name what develops into a seed
6	We do	13-21	Orchard evidence desk
7	We do	Within stage	Audit the missing step
8	We do	Within stage	What does the evidence prove?
9	Hinge	21-24	Which structure becomes a seed?
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A pollen delivery mystery

Invite predictions without requiring existing terminology. Explain that paper evidence gives the whole investigation.

2 Retrieval Recover the plant words

Valid part; new plant; no, reproduction makes offspring, whereas growth increases size/development of an individual.

3 I do I trace the correct delivery destination

Think aloud: I place my token on the anther, then move it to the stigma. A landing on a leaf would not complete this defined transfer. This selected diagram omits petals and exact scale.

4 I do I distinguish the event inside the ovule

Shared I do time. Think aloud: The transfer has already happened; now I describe a different event. The male gamete travels through a tube; the egg is in an ovule within the ovary. Advanced double fertilisation is omitted.

5 I do I name what develops into a seed

Shared I do time. Trace each independent arrow. Ordinary botanical fruits include pods; fruit here is a plant structure, not just a sweet food.

6 We do Orchard evidence desk

A shows transfer to stigma; D records pollen on leaf. Ask pupils to place or point to the appropriate destination on a diagram.

7 We do Audit the missing step

A pollination → B fertilisation → C seed/fruit development. A bee does not carry ready-made seeds into the ovary.

8 We do What does the evidence prove?

No; pollen transfer is seen, not gamete fusion. State that the record supports pollination but does not establish later success.

Reveal: The record shows pollen on a stigma. It does not by itself prove fertilisation or a mature seed.

9 Hinge Which structure becomes a seed?

A false pollen/seed confusion; B correct; C false leaves not seed precursors here.

A. The pollen grain simply changes into a seed. - Pollen supplies male gametes; the ovule develops into the seed.

B. A fertilised ovule develops into a seed. - This distinguishes reproductive cells from the structure that develops.

C. A leaf changes into every seed. - In this model, seeds develop from ovules inside the ovary.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

Transfer anther → stigma; gamete fusion; fertilised ovule. E4 on pupil page asks why pollen on leaf is insufficient.

Answers and assessment

R1-R3

R1 valid plant part. R2 a new plant in suitable conditions. R3 no; reproduction produces offspring, whereas growth develops an individual.

W1-W3

W1 A shows pollination; D has pollen on a leaf, not the required stigma. W2 A→B→C: transfer, fusion, development. W3 pollen on stigma is observed; successful fusion and later seed development are not established by that observation alone.

Hinge A-C

A incorrect: pollen does not become seed. B correct. C incorrect: ovules, not leaves, develop into seeds in this model.

S1-S5

S1 anther; stigma. S2 pollination. S3 fertilisation. S4 fertilised ovule. S5 pollen reaches stigma→gametes fuse→seed develops; any accurate arrow explanation.

N1-N4

N1 pollination: anther-to-stigma transfer; fertilisation: fusion of male/female gametes; seed formation: fertilised ovule develops. N2 pollen arrives at stigma, tube grows and delivers male gamete to egg cell in ovule. N3 leaf is not stigma. N4 transfer is evidenced, not later fertilisation or mature seed success.

T1-T4

T1 accurate connected sequence, with distinct transfer and fusion. T2 bees can transfer pollen, not plant seeds into ovaries; ovules develop into seeds and ovaries into fruit. T3 compatible pollen, tube growth/gamete delivery, fertilisation and suitable development conditions remain relevant; choose two without requiring technical detail. T4 separate dated records for pollen arrival on stigma and later seed formation, with a matched observation method; arrival supports pollination, later mature seeds support successful development in the sexual example. A record should not claim unobserved cellular detail.

E1-E4 / assessment

E1 transfer anther→stigma. E2 fusion of gametes. E3 ovule→seed, ovary→fruit. E4 incorrect destination. Secure: clear distinction of the three processes and correct structure development. Re-teach by assigning a different verb to each arrow. Reflection has no fixed choice.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This lesson follows ordinary sexual reproduction in flowering plants; it does not claim every plant seed in nature requires fertilisation. Asexual seed formation is a later teacher nuance.

The interactive uses the same constructed records as W1-W3. Paper pollen tokens provide a complete alternative without exposure to real pollen.

Access and responsive teaching

Supported

Order named stages using prompts and label the main route.

Standard

Explain the three processes and interpret an orchard evidence set.

Stretch

Evaluate a pollination guarantee and explain limits of the observations.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

Pollination is the same as fertilisation.

Pollination transfers pollen; fertilisation is gamete fusion later in the ovule.

Check: Ask where each process occurs.

A pollen grain simply becomes a seed.

A fertilised ovule develops into a seed; the pollen supplies male gametes.

Check: Ask which structure develops into the seed.

Every pollen grain landing anywhere on a plant makes a seed.

Pollen must reach a suitable stigma; further processes and suitable conditions are needed.

Check: Ask whether pollen on a leaf counts as pollination.

Word	Meaning
pollen	Grains that carry the male reproductive cells in flowering plants.
pollination	Transfer of pollen from anther to stigma.
stigma	The flower part that receives pollen.
ovule	A structure in the ovary containing an egg cell; it can develop into a seed.
fertilisation	Fusion of male and female gametes.
gamete	A reproductive cell such as an egg cell.

Scientific references

OpenStax Biology 2e: Pollination and fertilization

Checked September 2026. Pollination differs from gamete fusion; fertilised ovule develops into seed and ovary into fruit. College detail is selectively simplified for GROW.